For example, the following configuration was used to match all the dynamic elements in an application that used the Amazon API:

```
VCR.configure do |c|
  c.cassette_library_dir = 'test/vcr'
  c.hook_into :webmock
  c.ignore_localhost = true
  c.default_cassette_options = {
    :match_requests_on => [:method,
      VCR.request_matchers.uri_without_params(
        "Timestamp", "Signature", "AWSAccessKeyId", "AssociateTag")]
  }
end
```

The match_requests_on option is used to match on the combination of HTTP method and URI but with several parameters to the URI ignored.

The filter_sensitive_data option lets you keep passwords and the like from appearing in your VCR cassette. It allows you to use a block to grab the sensitive text from the HTTP response and replace it with a custom dummy string.

VCR has other options for more elaborate use. You can see the full list in the documentation.[6]

The World Is a Service

Once you get used to the idea of having adapters mediate access between your application and external services, it's not that far a leap to have adapters internally to mediate between different parts of your application. This approach is sometimes called *hexagonal architecture*, and many resources can be found online that describe hexagonal architecture as it applies to Rails.[7] At the same time you can find many resources online saying that hexagonal architecture is an awful idea. (David Heinemeier Hansson is a particularly vocal critic.)

You've taken baby steps in this direction by creating action objects such as CreatesProject, which are somewhat like adapters between the controller and the model. Many web frameworks use adapters between model objects and the database. Rails does not, but the pattern is not uncommon.

What You've Done

In this chapter you looked at a pattern for testing external services while still allowing the test suite to run quickly and consistently and allowing you to set up and test for error conditions. You used an adapter in your code to

6. https://relishapp.com/vcr/vcr/v/2-9-2/docs
7. https://medium.com/@vsavkin/hexagonal-architecture-for-rails-developers-8b1fee64a613